

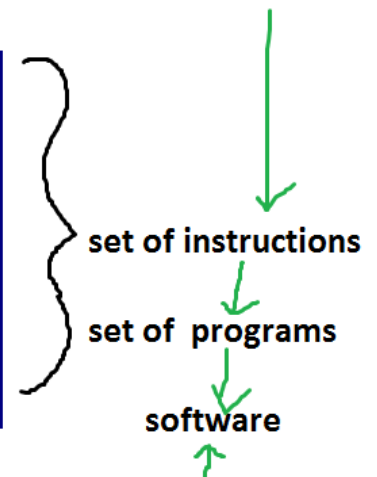
# INTRODUCTION TO C

C is a

1. High level / middle level programming language.
2. C is a compiler based programming language.
3. C follows procedure oriented programming structure [ POP's ]

c / c++ / java / py / .net - programming languages

```
/* source code / source program */
#include<stdio.h>
#include<conio.h>
void main()
{
    int i; /* variable */
    clrscr();
    for(i=1;i<=10000;i++)
    printf("%d ",i);
}
```



software engineer / programmer / developers / techies

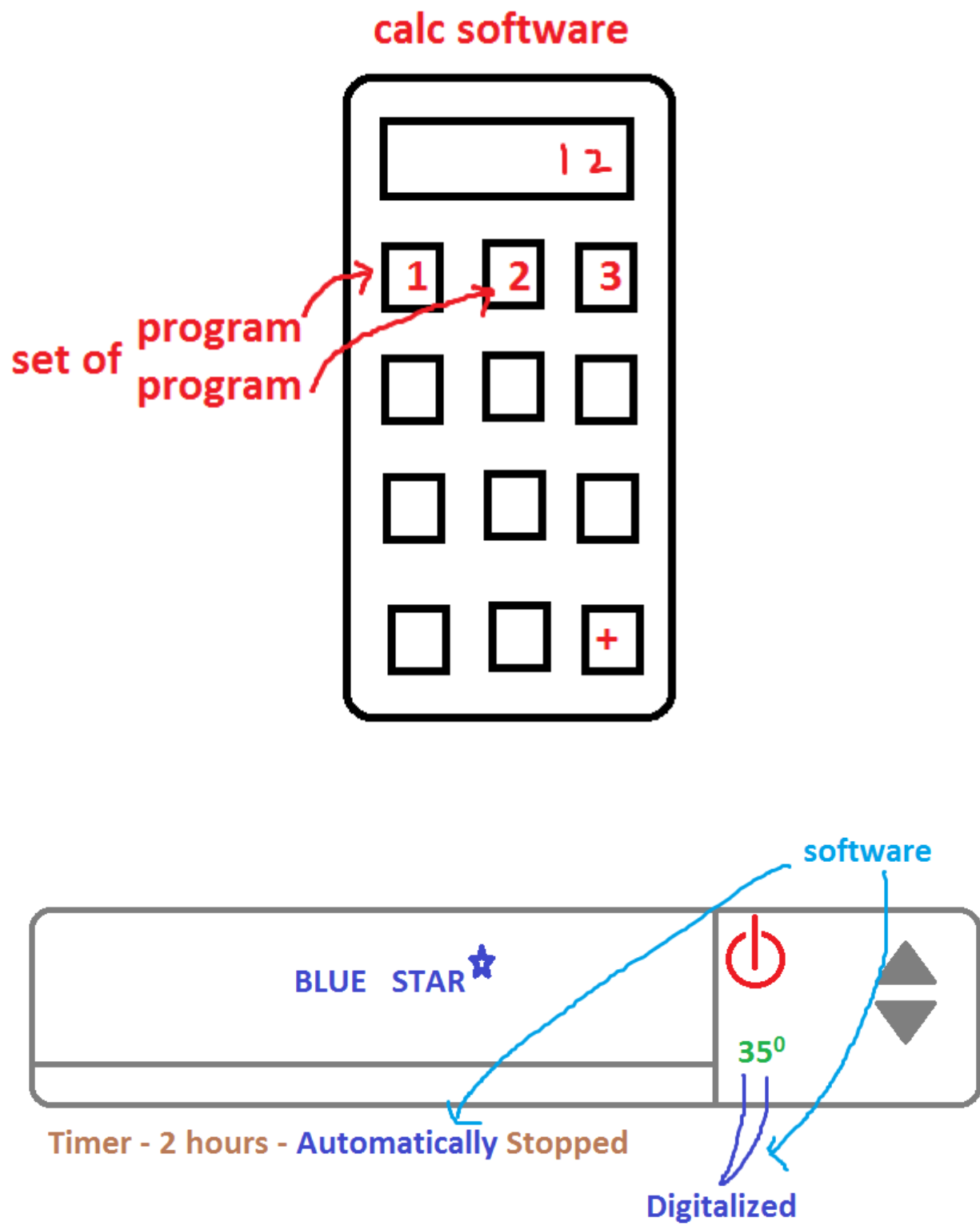
wipro / tcs / infosys / tech / microsoft / oracle - software companies

## What is a program?

Set of instructions is called program.

What is a software?

Set of programs is called software / As per IT industry software is a digitalized and automated process.



We are having 2 type of software.

## 1. System software

Eg: operating system, translators, device drivers

## 2. Application software

Eg: Whatsapp, insta, phonepe,....

## What is a language?

Generally the languages are used to communicate with others. We are using the human languages like telugu, English, hindi, Marathi etc are used to communicate with humans. But by using these human languages we can't communicate with machines.

For that we are using the **computer programming languages** like C / C++ / Java / Py / .net etc, which allows to write the programs [ software ] to communicate with devices.

Basically the computer languages divided into 3 types.

### 1. **Machine language**: Created with binary code

Eg: 1000111

### 2. **Low level/assembly language**: Created with English shortcuts called **MNEMONICS**.

Eg: sub, add,

Example:

Turbo C++ IDE

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ASS.CPP

```
// example for low level / assembly program
// example of adding two numbers
#include<stdio.h>
void main() {
    int a = 10, b = 20, c;

    asm {
        mov ax,a
        mov bx,b
        add ax,bx
        mov c,ax
    }

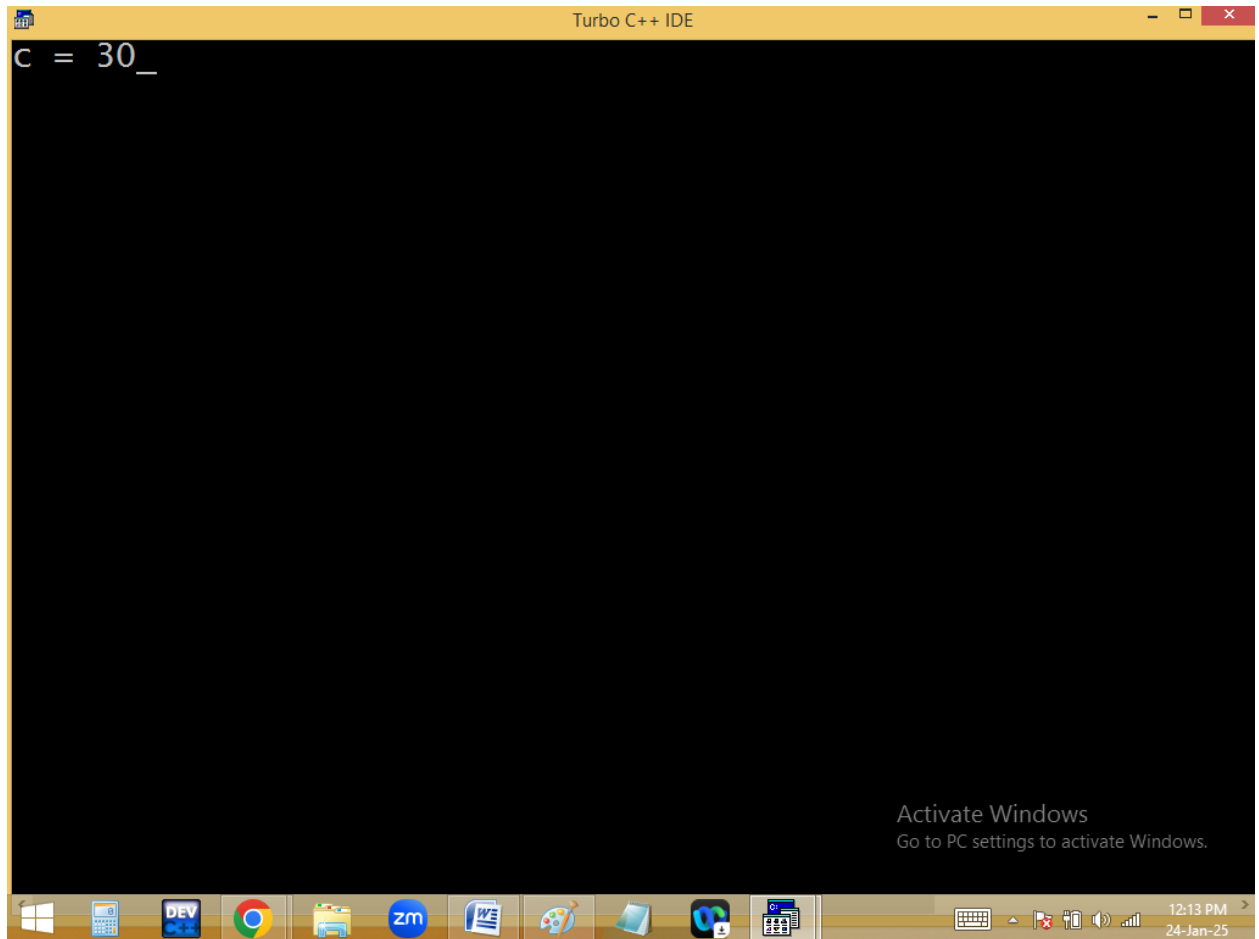
    printf("c = %d",c);
}
```

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Activate Windows  
Go to PC settings to activate Windows

F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile

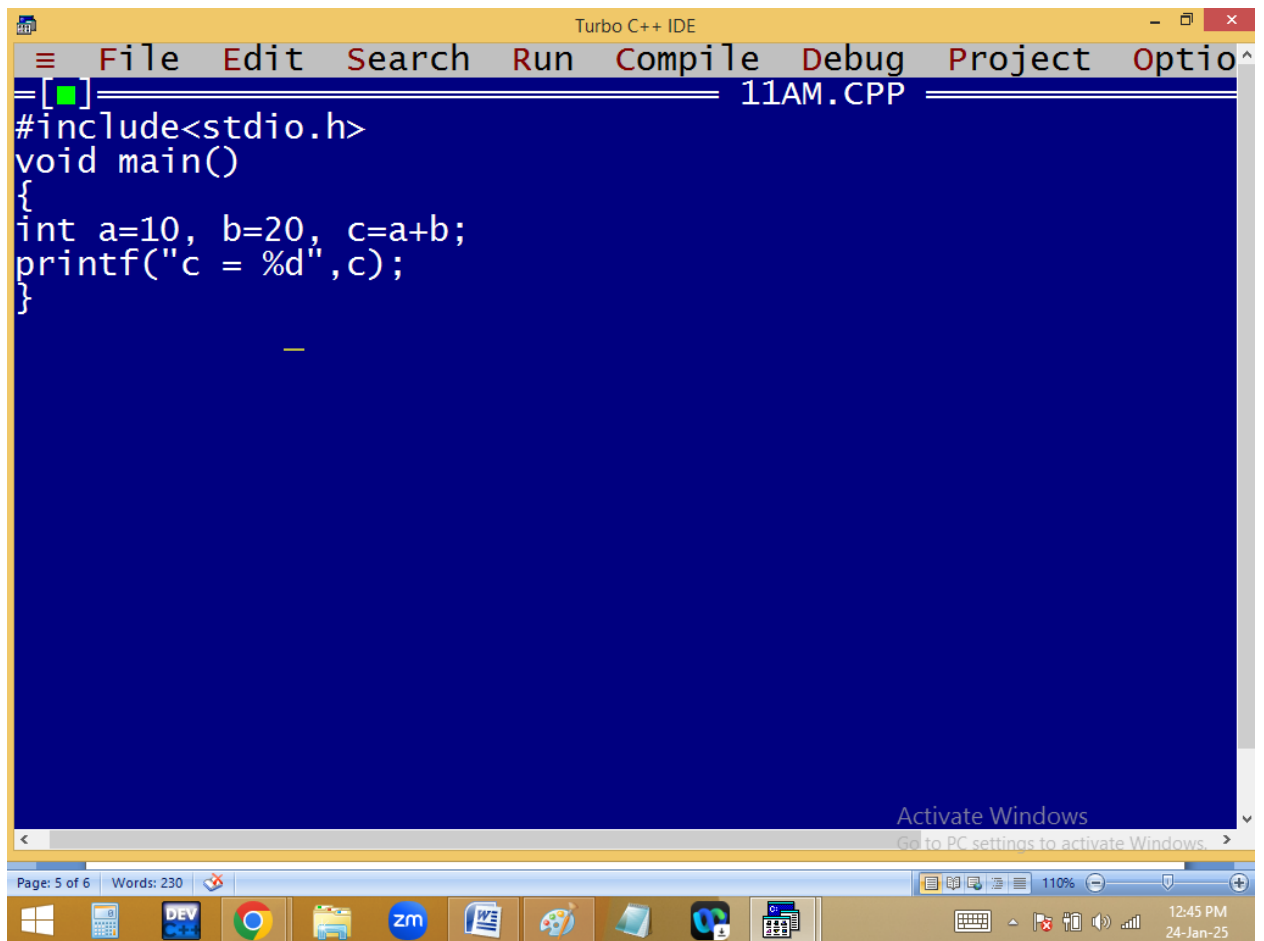
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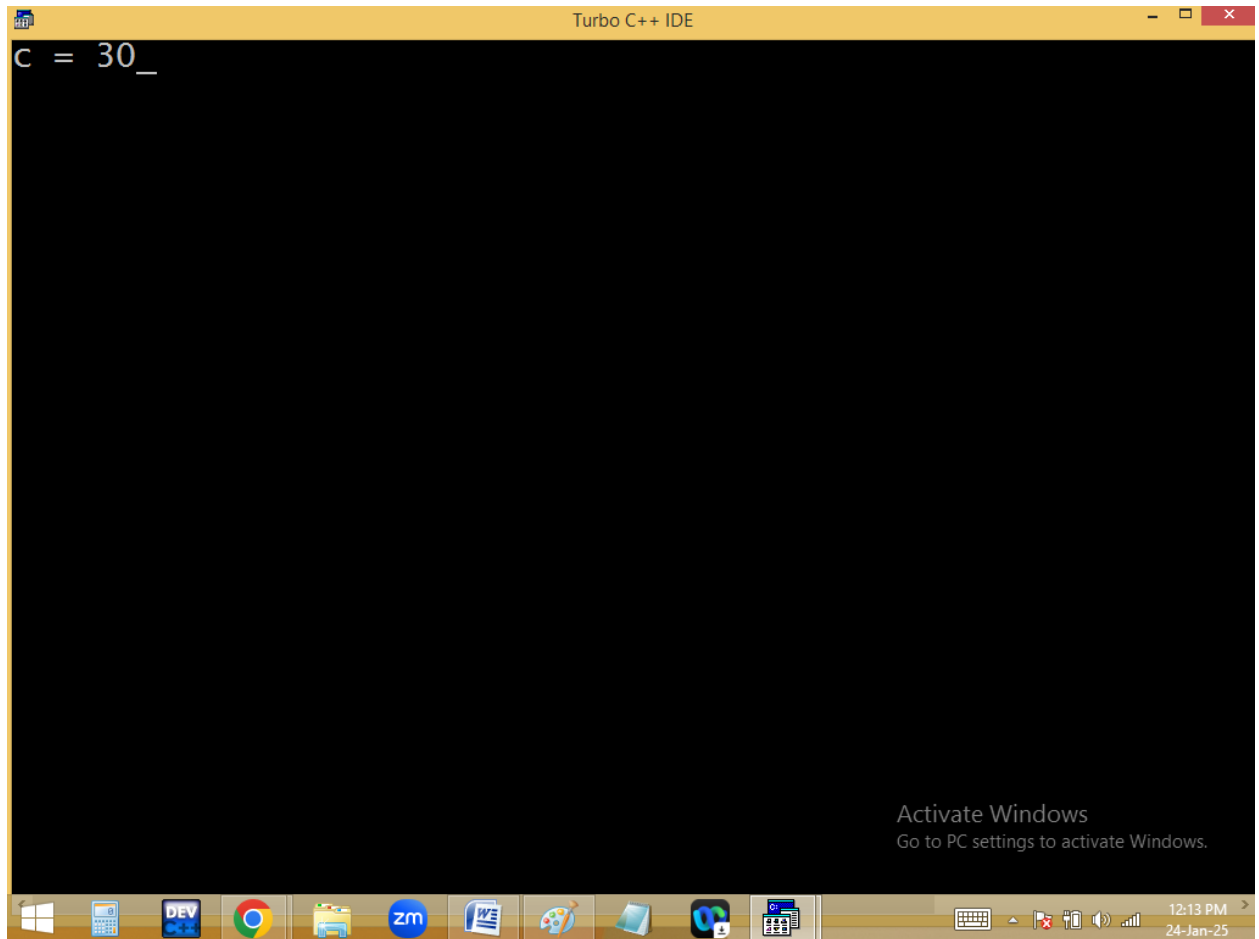


3. **High level language**: Created with simple English.

Eg: good morning, subject, addition,

Example:





C is a high level language with low level features. Hence it is also called middle level language.

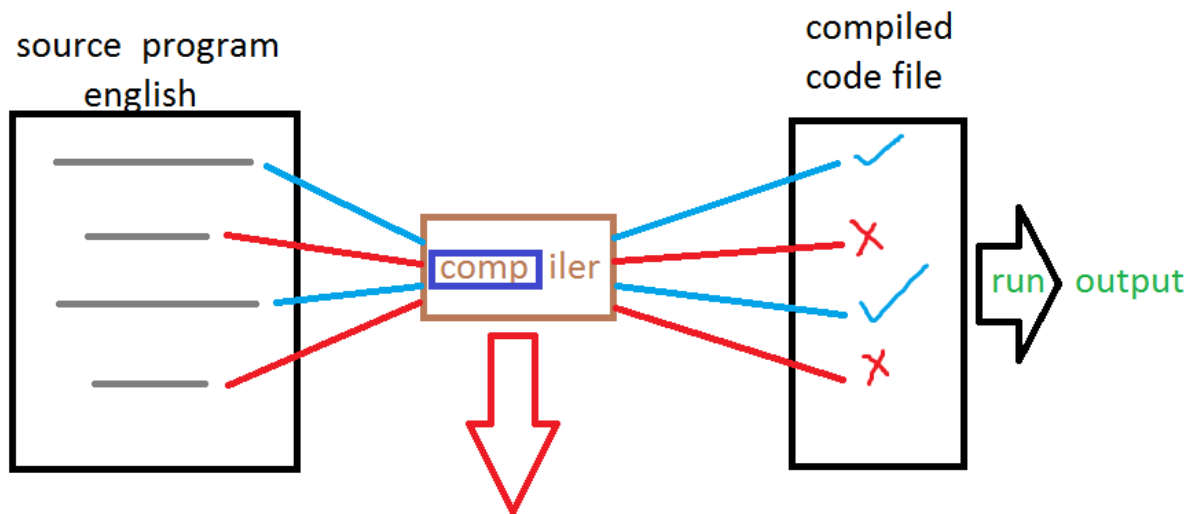
C high level features used to develop application software and low features used to develop system software. Hence C is a multi-purpose programming language.

## What is a translator?



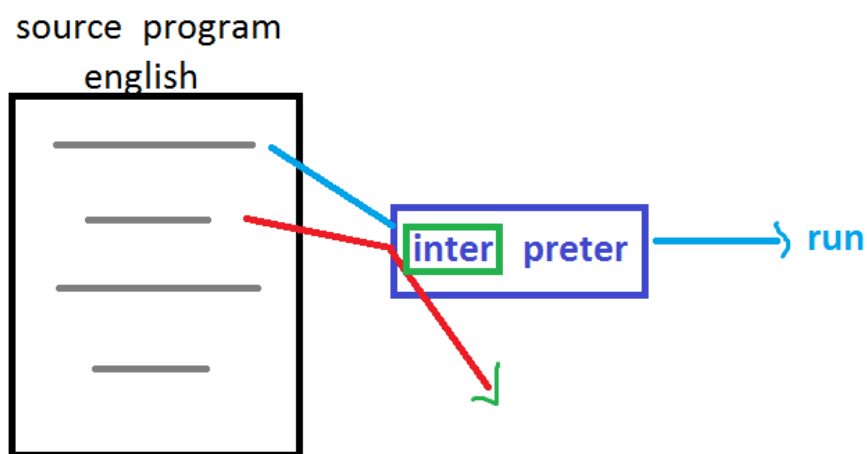
Compiler and interpreter used to convert our high level programs to binary code.

Compiler converts the total source code into binary code at once by leaving error lines.



After **completion** of total program it shows the errors

Interpreter converts **line by line**.



showing error in 2nd line [**middle**] and not checking remaining lines

Assembler is used to convert low level programs to binary code.

Assembler working style is similar to compiler.

**In c we are using compiler. Hence it is a compiler based programming language.**

**In Java / .Net / Py we are using both compiler and interpreter. Hence they are called compiler based interpreted languages.**

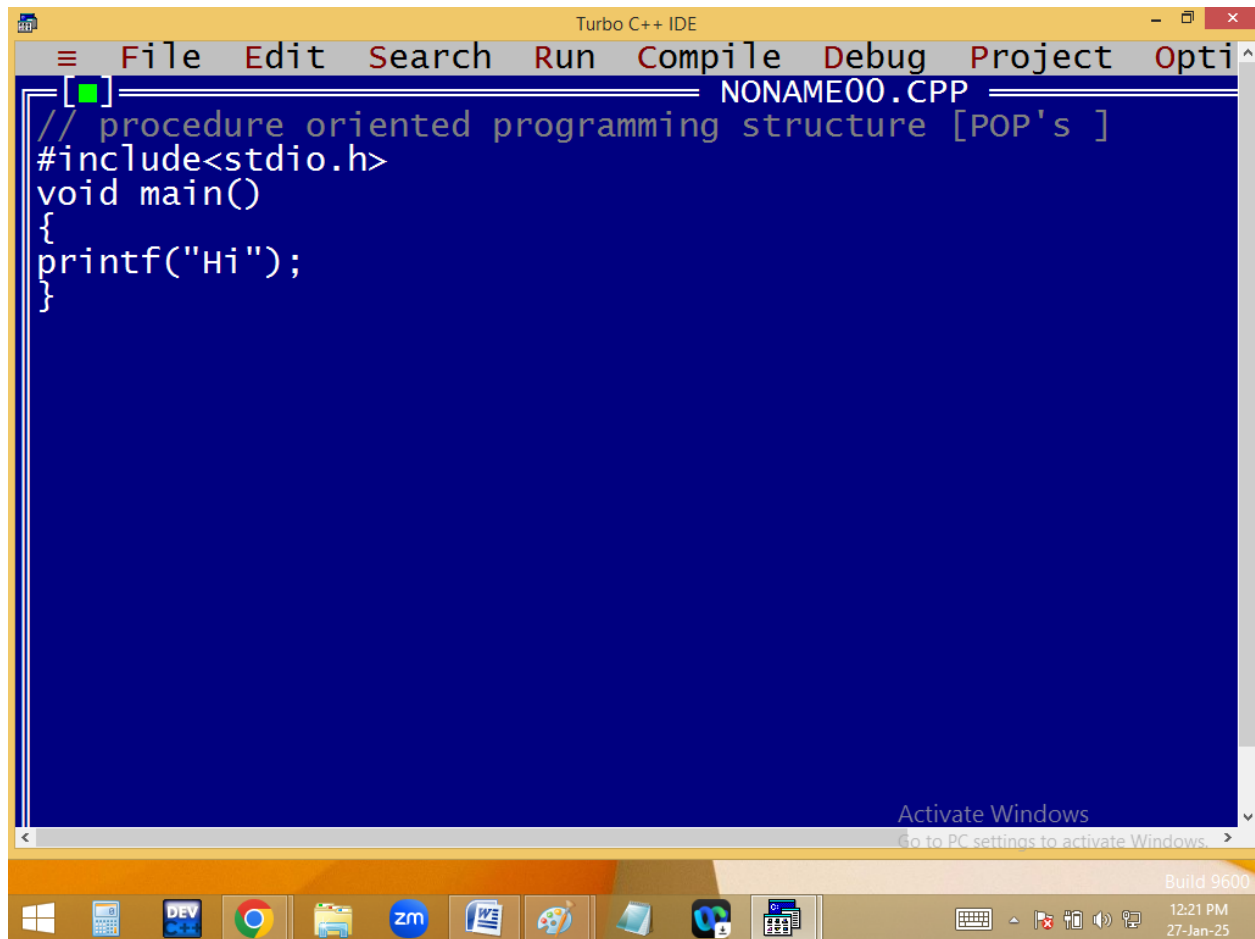
### **What is called programming paradigm?**

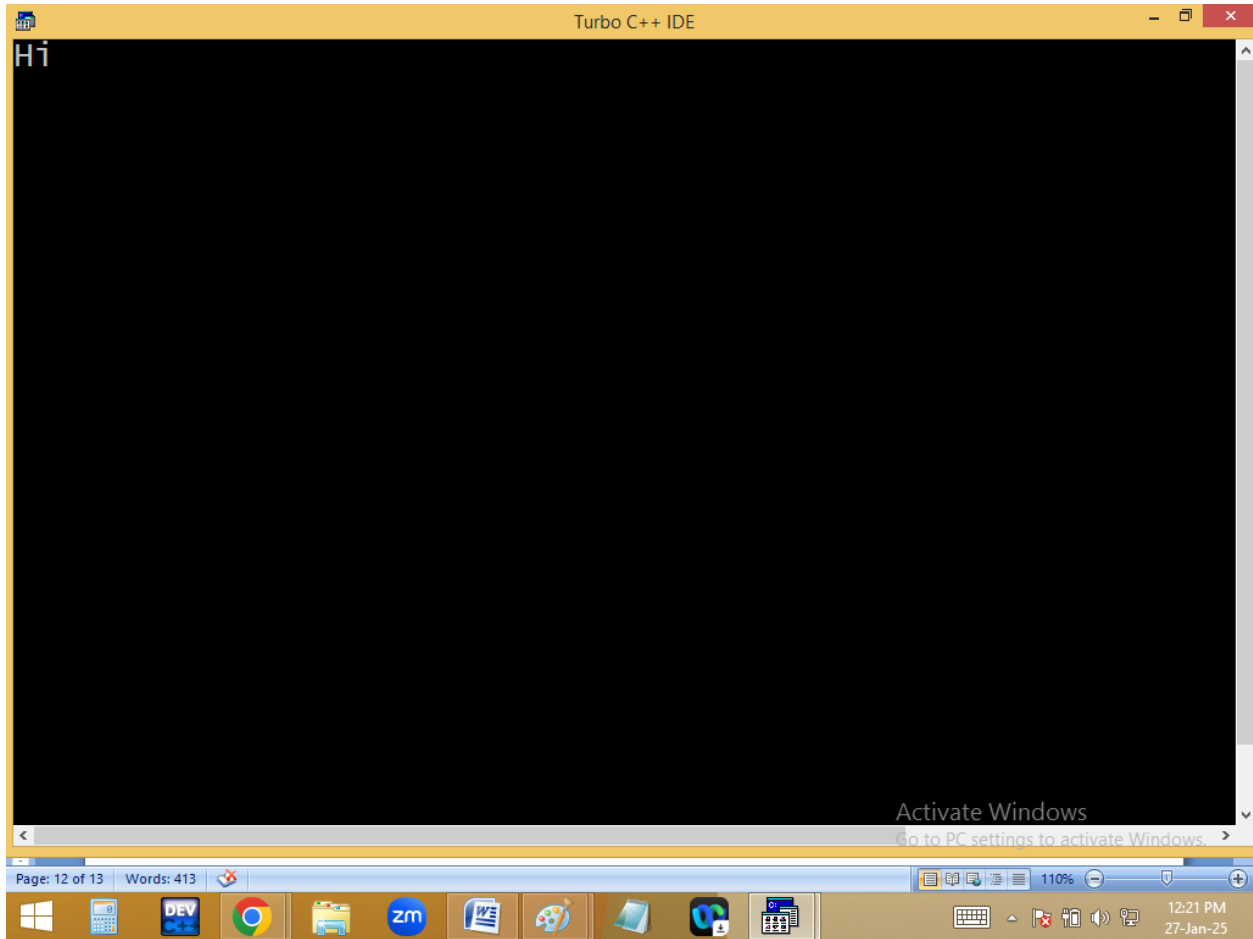
Every programming language is having certain rules and regulations with a particular programming architecture. As programmer we should have to follow these rules and structure, which is technically called programming paradigm.

**Before C language, the languages are using monolithic programming structure.**

**C is using Procedure oriented programming Structure**

**C++ and Python are using Procedure oriented and Object oriented programming Structures. Hence they are also called multi paradigm programming languages.**





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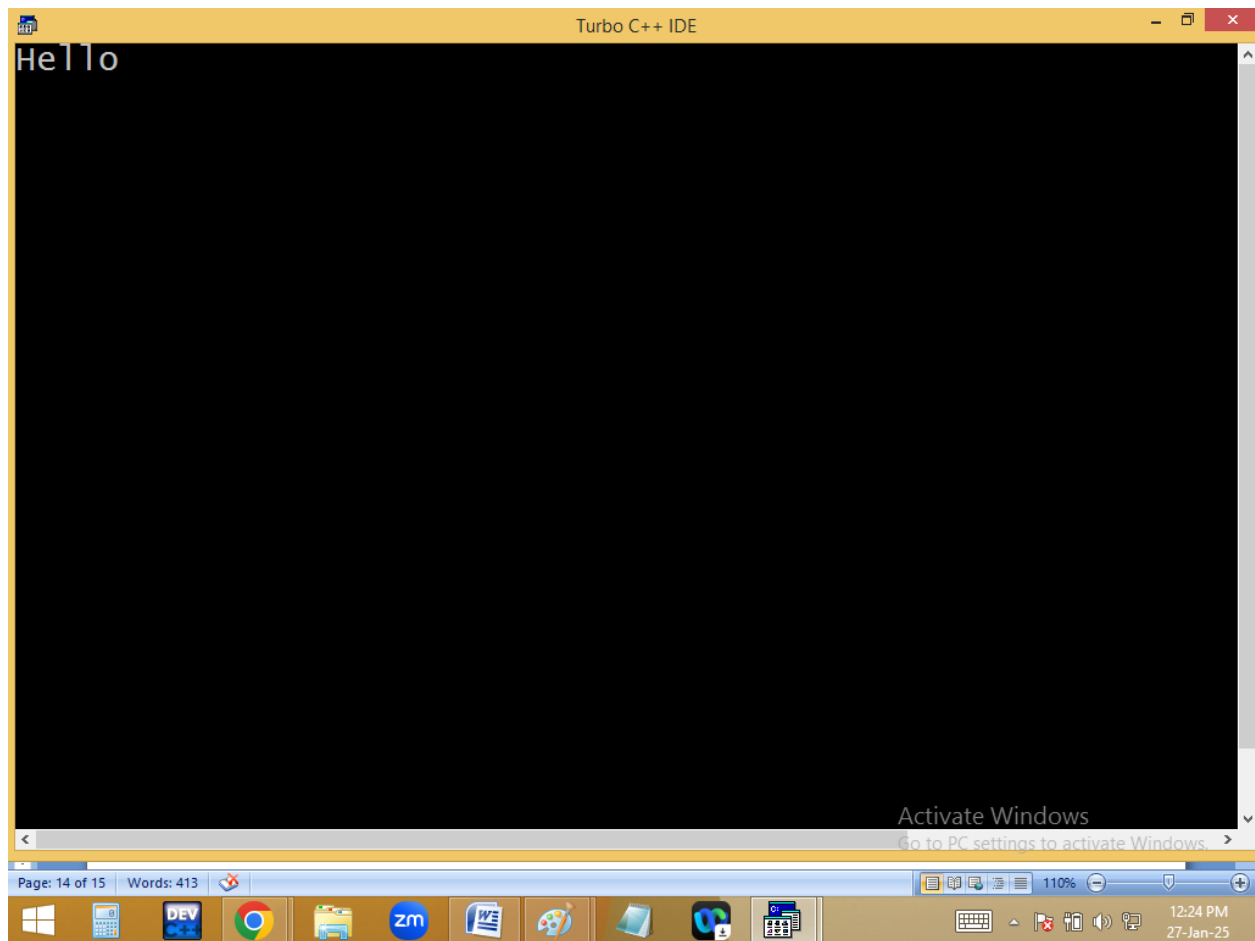
NONAME00.CPP

```
// Object oriented programming structure [POP's ]
#include<stdio.h>
#include<conio.h>
class test
{
public:
void show()
{
printf("Hello");
};
};
void main()
{
test t; // object
clrscr();
t.show();
}
```

Activate Windows  
Go to PC settings to activate Windows.

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Java and .net are using object oriented programming structure.

Modularity:

Simplicity

Reusability: Write once, use many times.

Efficiency: performance high.

```

/* Eng-source pro */_
#include<stdio.h>
void main()
{
printf("Good morning");
}

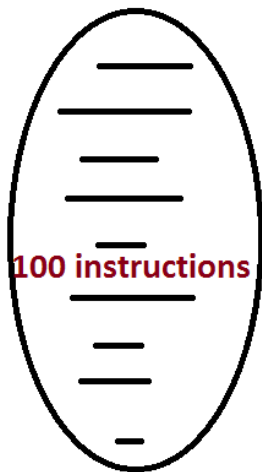
```



**Translators**  
**1. compiler**  
**2. interpreter**  
**3. assembler**

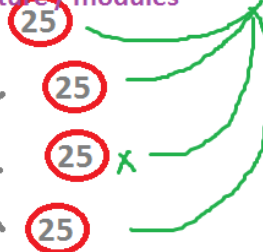
- 1. Checking errors**
  1. compile time
  2. runtime
  3. warnings
- 2. eng to binary**

program-source code  
**english**



**compiler**  
 Error checking  
 eng to binary

subprog/subroutines/procedures/functions /  
 structure / modules



procedure oriented programming  
 structure - POP's

**monolithic programming**

- dis: monolithic**
1. diff find error
  2. takes more memo
  3. performance low
  4. no reusability